

JEE MAIN REVISION MASTER: SOME BASIC CONCEPTS OF CHEMISTRY SYLLABUS REDUCED

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 40 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Some Basic Concepts Of Chemistry - Core Application	40 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Some Basic Concepts Of Chemistry - Core Application (Very High Yield — 40 PYQs)

Core Principle: Key problem pattern in Some Basic Concepts Of Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Match List - I with List - II. List - I Mass of substance List - II Number of atoms A. 1.8 mg water I. 2×10^{-4} times N_A ..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "How many grams of residue is obtained by heating 2.76 g of silver carbonate? (Given : Molar mass of C, O and Ag are 12, 16 and 108 g mol^{-1} ..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "The ratio of mass percentage (w/w) of C : H in a hydrocarbon is 12 : 1. It has two carbon atoms. The weight (in g) of CO_2 (g) formed when 3..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#3)

An oxide of iron contains 69.9% iron, its empirical formula, is : (Given : Molar mass of Fe and O are 56 and 16g mol⁻¹ respectively.)

- (A) FeO
- (B) Fe₂O₃ [Correct]
- (C) Fe₃O₄
- (D) FeO₃

Correct Answer: B

Solution Analysis: Let the oxide of iron contain 100 g of the compound. Then, Mass of Fe = 69.9 g Mass of O = 100 – 69.9 = 30.1 g Calculate moles of each element Moles of Fe = (69.9 / 56) = 1.248 Moles of O = (30.1 / 16) = 1.881 Find the simplest whole-number ratio Divide each by the smaller number (1.248): Ratio of Fe:O = (1.248 / 1.248) : (1.881 / 1.248) = 1 : 1.51 This ratio is approximately 1 : 1.5 . To remove the fraction, multiply both by 2 : Fe₂O₃ In Fe₂O₃ , Mass of Fe = 2 × 56 = 112 g Mass of O = 3 × 16 = 48 g Total mass = 160 g Percentag